
CUBE: A Unified Standard and Benchmark Suite for Evaluating Generalist Agents

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Abstract

1 The agentic benchmark landscape now exceeds 600 environments and is still
2 growing, yet each one ships its own provisioning logic, action space, tool
3 definitions, and evaluation harness. The resulting $N \times M$ integration tax blocks
4 cross-benchmark research, including the multi-environment RL post-training that
5 recent work identifies as the strongest path to generalist capability. We introduce
6 **CUBE** (Common Unified Benchmark Environments): a Python-first standard
7 for tasks, benchmarks, tools, and a three-tier resource lifecycle, with an optional
8 JSON-RPC interface for non-Python clients. Any runtime that drives the Task
9 gym API is a valid harness; AgentBeats and NeMo Gym already consume CUBE
10 benchmarks. The reference harness adds auto-provisioning across Docker, VM,
11 and live backends, a registry of 10 wrapped benchmarks across web, computer-use,
12 SWE, and terminal modalities, and a trajectory judge that attributes per-episode
13 failures to agent, tool, or benchmark causes via $K=3$ cross-judging with post-hoc
14 evidence verification. We evaluate 4 frontier models on 8 CUBEs with tool
15 stacks held constant per modality, exposing capability profiles that per-benchmark
16 leaderboards do not surface and pinpointing where failures live: roughly 82 %
17 trace to the LLM itself, with the rest concentrating on specific tool stacks and
18 on wrapper bugs that the judge surfaces and that we then fix in place. Because
19 a wrapped benchmark is only useful if it is the same benchmark, we measure
20 wrapper fidelity directly rather than inferring it: task-set identity against the
21 upstream index (368/368 and 152/152 instructions byte-identical for OSWorld
22 and Windows Agent Arena), evaluation provenance per CUBE, and oracle parity
23 against each benchmark’s own reference solution. Wrapping a new benchmark
24 reduces to a few typed classes plus metadata (median 508 lines across our corpus),
25 and a Claude skill turns even that into an afternoon of work.

26 1 Introduction

27 The number of agentic benchmarks has grown from a handful to over 600 in five years (Figure 1).
28 More than 100 new benchmarks were published in Q1 2026 alone, spanning web navigation, computer
29 use, software engineering, embodied control, scientific discovery, and a dozen other categories.¹ This
30 growth reflects genuine scientific demand: agentic evaluation is hard, domain coverage is uneven,
31 and researchers keep finding gaps that require new environments. The proliferation is a sign of health,
32 not pathology. The problem is that each benchmark arrives as an island, shipping its own scaffolding.
33 WebArena requires a set of web servers provisioned from a specific snapshot. SWE-bench requires
34 per-task Docker containers cloned from a repository matrix. OSWorld requires a pre-provisioned
35 virtual machine image with a running desktop. WorkArena requires a live ServiceNow instance.
36 Every benchmark defines its own action space, names its own tools, makes its own assumptions
37 about how an agent is called, and provides its own (often fragile) evaluation harness. A team that
38 wants to evaluate a single agent across five benchmarks pays five independent integration costs. The

¹ Corpus curated via deep search of arxiv listings; each entry requires an interactive environment, links to a published paper, and has been individually verified.

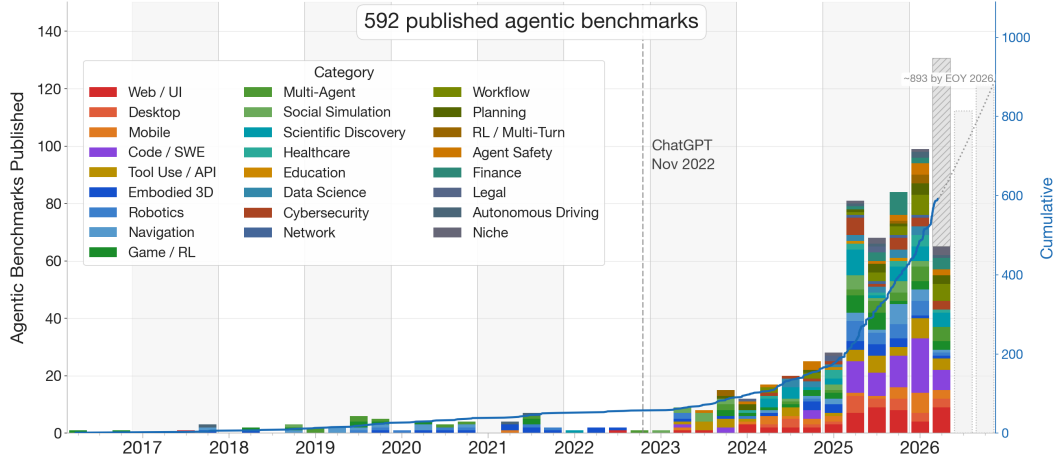


Figure 1: Agentic benchmarks published per quarter, 2016–2026, by category (cumulative count on right axis). Dashed line: ChatGPT (November 2022). Q2 2026 is $\approx 37\%$ complete; the gray hatched bar estimates remaining Q2 publications from a monthly linear trend since January 2023. Outlined bars and dotted projection line show Q3–Q4 2026 forecasts from the same model. See §5 for corpus details.

cost multiplies with models: N models on M benchmarks means $N \times M$ integration points. No wonder cross-benchmark results are rare and comparisons are almost always informal.

This fragmentation blocks more than evaluation. Recent work argues that future agentic systems should be designed for generality across diverse environments [Bandel et al., 2026], and large-scale reinforcement learning post-training has emerged as a primary route to capability gains in frontier models [Khatri et al., 2025]: every additional environment the model trains on adds a dimension of generalization it could not acquire from any one benchmark alone. With 600+ agentic benchmarks now in existence, the ideal training curriculum already exists on paper. But sampling from hundreds of environments at training time requires every benchmark to expose the same interface, the same resource lifecycle, and the same tool vocabulary. No such interface exists today. Every custom integration is a constraint on the training distribution, and a narrower distribution caps what can be learned. The integration tax on evaluation is the ceiling on generalist capability tomorrow.

We introduce **CUBE** (Common Unified Benchmark Environments), addressing this bottleneck through four contributions.

The CUBE Standard (Section 3): a Python-first set of abstractions for tasks, benchmarks, tools, and resources, with a three-tier resource lifecycle for transparent auto-provisioning across Docker, VM, and cloud backends, and an optional JSON-RPC interface for non-Python clients. The standard also introduces a composite benchmark class that subsets and combines existing CUBEs as a difficulty-calibrated cross-modality benchmark.

A reference harness (Section 4): although the standard is harness-agnostic and interoperates with AgentBeats and NeMo Gym, we provide a reference harness with parallel execution, OpenTelemetry tracing, trajectory logging for RL training data, and a trajectory judge that attributes per-episode failures to agent, tool, or benchmark causes, making cross-benchmark failure analysis tractable at scale.

A wrapped corpus and registry (Section 5): 10 CUBEs spanning web, computer use, software engineering, and terminal modalities, with automated compliance verification and a live registry. At the time of writing the registry includes a cube for MiniWoB [Shi et al., 2017], WebArena [Zhou et al., 2024], WorkArena [Drouin et al., 2024], BrowseComp [Wei et al., 2025], OSWorld [Xie et al., 2024], Windows Agent Arena [Bonatti et al., 2025], SWE-bench [Jimenez et al., 2024], SWE-bench Verified [Chowdhury et al., 2024], Terminal-Bench [Merrill et al., 2026], and DRBench [Abaskohi et al., 2025]; and with the release of our claude skill to wrap any benchmark within CUBE, we expect the registry to grow significantly.

70 **Cross-modality evaluation (Section 6):** a systematic evaluation of frontier models across web,
 71 CUA, and SWE tasks with tooling held constant per modality, exposing capability profiles that
 72 per-benchmark reports tend to miss.

73 Standardization enables research that is otherwise impractical. When tools are first-class objects
 74 in the standard, tool design becomes an independent variable: one browser tool drives WebArena,
 75 WorkArena, and MiniWoB, making tool ablations honest across benchmark families. When
 76 composites are a core abstraction, difficulty can be recalibrated as agents improve without
 77 re-engineering anything. When every benchmark speaks the same interface, curriculum training
 78 across hundreds of environments is a sampling-policy decision, not an integration project.

79 2 Related Work

80 **Agentic benchmarks.** The agent-evaluation landscape now spans hundreds of environments
 81 [Yehudai et al., 2025], including multi-environment benchmarks such as AgentBench [Liu et al.,
 82 2024], AppWorld [Trivedi et al., 2024], and τ -bench [Yao et al., 2025]; each individual benchmark
 83 CUBE wraps embeds strong infrastructure assumptions. WebArena [Zhou et al., 2024] requires
 84 a persistent shared server stack; WorkArena [Drouin et al., 2024] requires a live ServiceNow
 85 instance; OSWorld [Xie et al., 2024] and Windows Agent Arena [Bonatti et al., 2025] require VM
 86 snapshots with a running desktop; SWE-bench [Jimenez et al., 2024, Chowdhury et al., 2024]
 87 creates per-task Docker containers from a repository matrix; Terminal-Bench [Merrill et al., 2026]
 88 uses hermetic per-task containers; MiniWoB [Shi et al., 2017] runs inside a local browser. These
 89 incompatible resource models are the direct source of the $N \times M$ integration tax described in
 90 Section 1. BrowserGym [Le Sellier de Chezelles et al., 2025] demonstrates the value of a shared
 91 interface within the web modality; CUBE generalizes this approach across modalities and lifts the
 92 benchmark interface to include resource provisioning and tool configuration. HAL [Kapoor et al.,
 93 2026] surfaces a complementary problem: without controlled tooling, scores on the same benchmark
 94 are not comparable across papers. CUBE addresses this directly by making tools first-class objects
 95 in the standard, so the tool configuration is part of the benchmark’s declared interface.

96 **Platforms and orchestration.** Several systems address agentic evaluation and training from
 97 complementary angles. The METR Task Standard [METR, 2024] fixes a portable Docker/VM
 98 task-image convention with a small lifecycle of hooks; ~ 200 task families have been wrapped under
 99 it, and it is used across METR, UK AISI, and partner organizations. Inspect AI [UK AI Security
 100 Institute, 2024] is AISI’s evaluation orchestration framework, with 200+ pre-built evaluations,
 101 sandboxing, MCP [Anthropic, 2024] tool support, and bridges to coding agents. AgentGym [Xi et al.,
 102 2025] bundles 14 environments under a unified HTTP API, primarily as substrate for its AgentEvol
 103 training method rather than as a third-party-extensible standard. AgentBeats [AgentBeats Team and
 104 Berkeley RDI, 2025] proposes the Agentified Agent Assessment paradigm, in which benchmarks are
 105 realized as judge agents communicating over A2A and MCP, and operates a registry of judge agents
 106 alongside hosted competitions. NeMo Gym [NVIDIA, 2025] provides high-throughput RL training
 107 infrastructure with a microservice architecture and integrations into NeMo RL and OpenRLHF;
 108 OpenEnv [Meta PyTorch Team and Hugging Face, 2025] distributes Gymnasium-style environments
 109 through the HuggingFace Hub. PipelineRL [Piché et al., 2026] streams weight updates to inference
 110 workers to overlap generation and training on long-trajectory rollouts, and Harbor [Shaw, 2025]
 111 grew out of Terminal-Bench as a container-based post-training pipeline.

112 These systems operate at distinct layers: image conventions (METR), evaluation orchestration
 113 (Inspect), curated training suites (AgentGym), and RL execution (the rest). CUBE sits at the wrapping
 114 layer with a Python-native typed object model for tasks, benchmarks, tools, and the resource lifecycle,
 115 a JSON-RPC contract for non-Python clients, a registry for third-party contributions, and reference
 116 connectors that expose any CUBE benchmark as a NeMo Gym environment or an AgentBeats green
 117 agent.

118 Table 1 makes the comparison concrete. The axis that separates CUBE from the rest is the *resource*
 119 *lifecycle*: a per-task hermetic container is the only provisioning pattern most frameworks model,
 120 which structurally excludes benchmarks built on one shared server reused across tasks (WebArena,
 121 WorkArena) and benchmarks that need a running desktop (OSWorld, Windows Agent Arena). Harbor
 122 covers more benchmarks than we do and ships a stronger RL data path; its per-task container model
 123 cannot express those two patterns. The comparison also surfaces a practice we adopt from Harbor

Table 1: Frameworks for agentic benchmark execution, on the axes that decide which benchmarks a framework can host. *Prov.* lists the resource patterns modelled: **C** per-task container, **S** shared service reused across tasks, **V** VM snapshot with a running desktop. *Tools swap* means the action space is a declared, replaceable object rather than fixed by the benchmark or by the evaluated agent’s binary. Counts are as of the sources we inspected (May 2026); environments and adapters are not equivalent units across projects, so we also give each framework’s wrapping unit.

Framework	Envs	Modalities W / G / S	Prov.	Tools swap	Non-Py client	Registry + CI	RL path	Wrapping unit
CUBE (ours)	10	✓✓✓	C S V	✓	✓	✓	~	Typed Python classes
Harbor [Shaw, 2025]	~65	××✓	C	~	✓	✓	✓	<code>task.toml</code> + Dockerfile
NeMo Gym [NVIDIA, 2025]	40+	××✓	C	~	✓	×	✓	FastAPI microservice
OpenEnv [Meta PyTorch Team and Hugging Face, 2025]	~20	~×✓	C	×	✓	~	✓	Docker + <code>EnvClient</code>
AgentBeats [AgentBeats Team and Berkeley RDI, 2025]	38+	✓~✓	—	✓	✓	✓	~	A2A “green agent”
Inspect AI [UK AI Security Institute, 2024]	200+	~×✓	C	✓	×	×	×	Task + solver + scorer
METR standard [METR, 2024]	~200	××✓	C V	×	×	×	×	Task family image
BrowserGym [Le Sellier de Chezelles et al., 2025]	~10	✓××	C S	✓	×	×	×	Gym env (web only)
AgentGym [Xi et al., 2025]	14	~×~	C	×	✓	×	✓	HTTP env server

rather than claim over it: Harbor requires every adapter to pass a quantified parity experiment against the original benchmark before it enters the registry. We report parity measurements for our own corpus in §6.2 and add a parity record to the registry contract.

RL post-training over agentic environments. Reinforcement learning post-training is now central to frontier LLM development and follows predictable scaling laws [Khatri et al., 2025]; recent position work argues that the natural endpoint is generalist agentic systems trained and evaluated across diverse environments [Bandel et al., 2026]. Prior work on multi-environment RL for agents has been limited to bespoke per-project integrations, with custom reset and reward logic per environment and no shared tool vocabulary. A standard task interface, resource lifecycle, and portable tool set are the precondition layer that makes large-scale curriculum training across heterogeneous benchmarks tractable rather than a per-project engineering effort.

3 The CUBE Standard

The CUBE standard serves two audiences. A *downstream user* (a harness, a training platform, or a researcher) needs to run any benchmark and provision its resources through a single uniform interface, without writing benchmark-specific code. A *benchmark author* needs a clear contract so that a single CUBE compliant implementation reaches all downstream users at once. The standard is harness-agnostic and ships as a pure Python library.² A central design principle is the *config/make split*: every runtime object (benchmark, task, tool) has a serializable config counterpart that crosses process boundaries safely; calling `.make()` on the config returns the live object that holds a dynamic state and is never serialized.

3.1 Running a CUBE

Running a CUBE is config-first. A downstream user assembles an experiment by composing serializable Python objects: a tool config, a benchmark config, an infra config, and an agent config. The framework separates shared setup from per-task execution: calling `.make(infra)` on the benchmark config provisions shared resources (servers, containers, VMs) once; the resulting task configs are lightweight serializable records that travel to Ray workers, which access or instantiate live environments locally based on their needs. Only serializable handles, (endpoints, IDs, infra config) cross to worker processes via a `runtime_context` dict, never live environments. Workers access live clients locally from those handles. The episode loop (right box) is gym-style: observe, act, repeat until terminal. The key addition over vanilla Gymnasium [Towers et al., 2024] is that `evaluate()` is decoupled from `step()` and can be called at any point independently of action frequency, which matters when agent and evaluator run on different schedules.

²Anonymous repository: <https://anonymous.4open.science/r/cube-standard-123>

Harness side: configure, iterate, dispatch

```
# serializable configs defines the experiment
tool_cfg = BrowsergymConfig(use_screenshot=True)
wav_cfg = WebArenaVerifiedConfig(tool_config=tool_cfg)
agent_cfg = MyAgentConfig(llm_config=...)
infra_cfg = AzureInfraConfig() # or AWS, GCP, local

@ray.remote
def run(task_cfg, runtime_ctx, agent_cfg):
    # right box -->
    return run_episode(task_cfg, runtime_ctx, agent_cfg)

# .make() provisions shared resources
# leaving the 'with' block tears them down
with wav_cfg.make(infra_cfg) as bench:
    rc = bench.runtime_context # endpoints, IDs, infra
    task_cfgs = bench.config.get_task_configs()
    ray.get([
        run.remote(tc, rc, agent_cfg) for tc in task_cfgs])
```

Gym-like Episode loop: one task, one agent

```
# evaluation running on a remote process
def run_episode(task_cfg, runtime_ctx, agent_cfg):
    task = task_cfg.make(runtime_ctx)
    agent = agent_cfg.make(task.action_set)
    obs, _ = task.reset()
    done = False
    while not done:
        act = agent.act(obs)
        out = task.step(act)
        obs, done = out.obs, out.done
    return task.evaluate()
```

Resource provisioning. A benchmark declares its infrastructure requirements as a list of `ResourceConfig`: the base image its containers or VMs need, any storage volumes, and setup scripts. When an experiment starts, the harness resolves these against the user’s `InfraConfig`, downloads the original benchmark artifacts, and builds or registers a benchmark-ready image on the target provider (local Docker, Azure, AWS, or others). Benchmark authors write their `ResourceConfig` once; users point at any supported infra target and benchmark code does not change. Three structurally different provisioning patterns appear across the corpus: (i) WebArena needs a shared server started once per run: the author starts it in `Benchmark._setup()` and passes the URL to tasks via `RuntimeContext`; (ii) SWE-bench needs a fresh container per task: the harness launches one via `runtime_context['infra']` when the task is constructed. (iii) OSWorld needs a pre-built VM image: the author declares a `VMResourceConfig`; the harness builds it on first use and caches it.

Composite benchmarks and RPC interface. `CompositeBenchmarkConfig` composes any number of existing benchmarks into a single standards-compliant one. Task IDs are prefixed by sub-benchmark name; routing is transparent to callers. Provisioning, RPC server, compliance verification, and registry metadata are all inherited. Composites nest freely. The standard also auto-generates a JSON-RPC 2.0 server with MCP-compatible endpoints from any `Benchmark` subclass, so any MCP client drives a CUBE task without an adapter layer (see Appendix A).

3.2 Wrapping a Benchmark

Every CUBE must provide a default tool that defines the agent’s action space: the tool is the benchmark’s public interface for any downstream agent or training pipeline, so tool selection is the first authoring decision. The standard ships reusable reference tools for each major modality. A BrowserGym-compatible browser tool drives web benchmarks (WebArena, WorkArena, MiniWoB); a desktop tool built on accessibility trees and PyAutoGUI drives CUA benchmarks (OSWorld, Windows Agent Arena); terminal tools drive SWE and scripted benchmarks. For benchmarks where tool design is itself a research variable, the separation between tool and task means a benchmark ships a sensible default while staying configurable: swapping the tool for the same benchmark produces directly comparable scores because the environment is held constant. Once a tool is chosen, authoring reduces to three steps: (1) declare benchmark metadata, task metadata and resources; (2) implement `reset()` and `evaluate()` on the `Task` subclass, and (3) ship a debug suite.

Tools. Custom tools follow the `@tool_action` pattern: the decorator derives the full action schema from the method signature and docstring, so no separate specification is needed. `ToolConfig.make(container)` instantiates the tool after its container is live; switching to a different backend means swapping the `InfraConfig` passed to `.make(infra)`; tool, task and benchmark code are unchanged.

Metadata and task execution info. A benchmark declares `BenchmarkMetadata` (name, version, description, task count, reset isolation mode) and per-task `TaskMetadata` (id, split, container config). Heavy per-task data (patches, archives, long problem statements, binaries) lives on a `TaskExecutionInfo` subclass populated lazily on workers, keeping initialization flat at scale. `BenchmarkConfig.install()` writes that data to a per-task cache (`$HOME/.cube/...`) once per worker; later runs read it from disk.

197 **Evaluation.** `task.evaluate()` is the benchmark author’s primary scientific contribution.
198 It returns a reward in $[0, 1]$ with a structured info dict. The base class calls it at episode end,
199 but the harness can also call it independently to support partial-credit scoring or judge-agent
200 analysis. Privileged evaluation data (ground-truth patches, evaluator scripts, internal state) lives
201 in the `task.execution_info` field, kept off the agent’s observation and tool surface; an opt-in
202 `task.get_privileged_info()` hook exposes it to judge agents or debug clients when needed.

203 **Debug suite and robustness.** Every CUBE ships two LLM-free contracts that the stan-
204 dard requires and the compliance check verifies. `get_debug_benchmark()` returns a
205 `BenchmarkConfig` pre-filtered to a small set of tasks with deterministic, known-correct so-
206 lutions. `make_debug_agent(task_id)` returns a scripted agent guaranteed to reach reward 1.0 on
207 those tasks without calling any language model. Running the debug suite against real infrastructure
208 proves that the provisioning pipeline, tool execution, and evaluation logic all work end-to-end, with
209 no model-in-the-loop ambiguity. The registry runs this suite post-merge whenever an entry changes;
210 the result feeds the entry’s status badge on the registry site. Locally, `cube test <name>` runs the
211 same suite, so authors catch broken environments before pushing.

212 **Authoring tools.** The standard ships with layer-by-layer specifications in `openspec/specs/`; each
213 spec includes Invariants and Gotchas sections that enumerate the most common authoring mistakes.
214 Two Claude Code skills exploit these specifications directly. `/new-cube` generates a complete
215 CUBE package from a benchmark description: task subclass, tool subclass, benchmark subclass,
216 debug suite, serializable configs, and `pyproject.toml`, all consistent with the standard’s invariants.
217 `/review-cube` audits an existing implementation against those invariants and reports violations
218 before the registry’s compliance check runs. These skills are the primary onboarding path for new
219 contributors and reduce the typical wrapping effort for an existing benchmark from days to hours.

220 4 A Reference Harness

221 The CUBE standard is harness-agnostic: any runtime that can instantiate a `BenchmarkConfig`
222 and drive the Task gym API is a valid harness. AgentBeats and NeMo Gym both consume CUBE
223 benchmarks directly, and a reference NeMo Gym connector in `cube-standard` exposes any
224 benchmark as a Gymnasium environment for RL frameworks in roughly 100 lines³. We provide
225 a reference harness that runs the full stack end-to-end and serves as the execution engine for the
226 experiments in Section 6. Adopting CUBE does not require adopting our harness. The harness
227 contributes three pieces: a generalist agent, a Ray-based experiment runner with trajectory logging,
228 and a trajectory judge that automates per-episode failure analysis.

229 **Generalist agent.** Genny, the agent used for the experiments in Section 6⁴, consumes any
230 CUBE-compliant action set without per-benchmark code and offers two cache-friendly context
231 modes. In *growing-history* mode, each step appends one (observation, action) pair to a stable
232 cacheable prefix; we use this for SWE agents, where tool-call results themselves carry the state
233 the agent needs to retain. In *rolling-summary* mode, a separate summarizer LLM produces a
234 per-step summary stored as its own message; the prompt keeps the accumulated summaries plus
235 the current observation while older raw observations are dropped, so the prefix cache extends cleanly
236 across steps. This mode covers UI-based agents (web, CUA), where retaining every screenshot or
237 accessibility tree would exhaust the context window within a few steps.

238 **Experiment runner.** The runner schedules episodes with Ray. Each worker receives a serialized
239 Episode (containing `TaskConfig`, `AgentConfig`, and the benchmark’s `runtime_context`), calls
240 `make()` locally to instantiate the live task, tool, and agent, runs the episode, and ships a Trajectory
241 back to the coordinator. No live object crosses a process boundary. The Benchmark itself stays
242 on the driver. The coordinator enforces per-step timeouts and retries transient failures (FAILED,
243 CANCELLED, STALE), tracking each episode through QUEUED → RUNNING → COMPLETED/FAILED.
244 Parallelism is bounded by infra capacity and the Ray worker pool.

245 Trajectories are persisted to a per-episode file store that feeds RL post-training pipelines, and
246 LiteLLM token counts give the cost-per-task figures used in Section 6.

³<https://anonymous.4open.science/r/cube-standard-123/src/cube/integrations/nemogym.py>

⁴https://anonymous.4open.science/r/cube-harness-123/src/cube_harness/agents/geny.py

4.1 Trajectory judge

Binary pass/fail tells you how often an agent fails, not why. With experiments running hundreds of episodes across multiple modalities, manual failure triage does not scale. As a remedy, the harness ships a per-episode judge that runs an external coding agent (Claude Code via `claude-agent-sdk`) over the recorded trajectory and emits a structured `JudgeOutput`: narrative analysis, outcome, primary blame category with confidence, additional blames, supporting evidence, and a failure hypothesis. Beyond diagnosing agent limitations, the blame distribution serves a second role: it validates wrapper faithfulness. A low benchmark-side blame rate (`env_failure`, `eval_brittle`, `task_unclear`) is evidence that the CUBE wrappers are not introducing systematic evaluation artifacts.

LLM-as-judge [Zheng et al., 2023, Gu et al., 2024] remains unreliable in general: judges underperform on objectively-correct response comparisons [Tan et al., 2025], hallucinate evidence and miscalibrate confidence [Jung et al., 2025], and naive judges systematically misclassify agent-trajectory outcomes [Lù et al., 2025]. Closer to our setting, MAST [Cemri et al., 2025] derived a 14-mode failure taxonomy for multi-agent traces with $\kappa=0.88$ agreement; the CUBE judge takes the same evidence-grounded approach at the single-agent level, with a 3-bucket attribution into agent, tool, or benchmark causes designed for use as a corpus-wide faithfulness signal across a 10-benchmark suite.

Grounded inputs. Per experiment, an Opus-class agent runs once (turn budget 80) to assemble a *context pack*: pointers and summaries for the agent prompt, tool definitions, task description, reward function, and infra entrypoints. A Sonnet-class judge then runs per episode (turn budget 40) with the context pack and the trajectory directory, invoked through `claude-agent-sdk`. Crucially, the judge reads files via its own tool loop rather than receiving a stuffed prompt, so attributions are grounded in real source: it can open a specific step’s accessibility tree or read the actual reward function.

Constrained outputs and reliability checks. The judge selects the *primary blame* from a fixed 10-category taxonomy (Appendix D) and reports `none` rather than inventing a cause; any non-`none` blame must be supported by step-indexed verbatim quotes pulled from the trajectory, and confidence is reported on a 0–5 scale. Two reliability layers sit on top. First, *post-hoc evidence verification*: every quote is checked against the persisted step file, mismatches (literal or fuzzy) trigger a confidence downgrade and catch the most common hallucination mode in our setting (synthesising a plausible-looking quote). Second, *cross-judging with $K=3$* : independent judges vote, the modal blame is reported, and quantitative aggregation is restricted to judgements with at least two-of-three agreement after verification. In practice $\geq 97\%$ of episodes reach majority agreement on the primary blame after this filter (Appendix F); transcripts are persisted with each episode record, leaving the remaining disputed cases auditable.

5 Benchmark Corpus and Registry

Our initial corpus spans web automation (WorkArena, WebArena-Verified, MiniWoB, BrowseComp), computer use (OSWorld, Windows Agent Arena), software engineering and terminal (SWE-bench Verified, SWE-bench Live, Terminal-Bench), and enterprise research (DRBench), totalling 5,642 tasks across four structurally distinct provisioning models (local browser, shared multi-service Docker stack, per-task Docker container, virtual machine, and live external service). The set was assembled to stress the standard from every direction: each modality forces a different combination of action space, infrastructure, and evaluation logic, and covering them in one corpus is what surfaced the abstractions in Sections 3 and 4. For the experiments in Section 6 we draw from the eight CUBEs in the Web, CUA, and SWE blocks; BrowseComp and DRBench are omitted to keep compute tractable. Per-CUBE task counts and backend details are tabulated in Appendix B.

A registry for a decentralized corpus. Any author can publish a CUBE in their own repository and ship it through any Python index. The registry is a lightweight file-based metadata index, GitHub-orchestrated rather than a hosted service⁵. Each entry is a small YAML record (name, version, authors, license, package link) submitted via pull request; the registry stores no benchmark code, only the pointer to a repository implementing that specific CUBE.

Automated compliance on submission. Submitting a CUBE triggers GitHub Actions compliance checks. A *quick check* (schema validation, package import, action introspection) runs on every PR in roughly two minutes. A *full check* post-merge executes the debug agent on the debug task set; a CUBE

⁵registry URL kept private during reviewing

is compliant only if the agent reaches reward 1.0 on every debug task. Passing both earns a compliance badge; failing the full check flags the entry degraded until a fix is submitted. This lets the registry accept third-party CUBEs at scale without trusting any individual submitter. Following Harbor [Shaw, 2025], which requires every adapter to pass a quantified parity experiment before acceptance, an entry also carries a *parity record*: upstream commit, provenance class, and oracle result (§6.2). The debug suite proves a CUBE *runs*; the parity record is what argues it is the *same benchmark*.

6 Experiments

Our experiments answer four questions. First, and most critically for an evaluation suite: are the CUBE wrappers faithful implementations of the underlying benchmarks (§6.2)? Second, can a single generalist agent configuration produce useful baselines across the corpus (§6.3)? Third, where do its failures originate: the LLM, the agent scaffolding, the tool stack, or the wrapped benchmark itself (§6.4)? Fourth, do those conclusions survive a change of agent scaffold (§6.5)?

6.1 Setup

Models. We evaluate four frontier models spanning two providers, with a small/large split per provider: Anthropic Claude Haiku and Claude Sonnet, OpenAI GPT-5.4-mini and GPT-5.4. All calls go through the harness’s LiteLLM gateway with default sampling parameters.

Agent. We use Genny (§4) as the single generalist agent across the corpus. Genny operates in two modes selectable per benchmark: rolling summaries for UI-based tasks (web, CUA), where retaining every screenshot or accessibility tree would saturate the context window; growing history for SWE and terminal tasks, where tool-call results carry the state the agent needs to retain. We do not tune a per-benchmark prompt; the only benchmark-aware choices are the context-management mode and the registered tool stack.

Tool stacks. One tool stack per modality, held constant across all benchmarks in that modality. Web: BrowserGym’s high-level action set. CUA: accessibility-tree input with PyAutoGUI for action output. SWE: a bash tool with structured stdout/stderr capture. Holding tooling constant per modality is what lets per-modality results be compared directly across benchmarks.

Benchmark subsets. We use full sets where tractable. WorkArena is restricted to L1 and L2 (L3 omitted); SWE-bench Live [Zhang et al., 2025] to *lite-gold-226*, the lite-split tasks whose gold patches pass evaluation verbatim; BrowseComp and DRBench are excluded. Subset sizes appear in Table 2.

Reporting. For each (model, CUBE) pair we report pass rate as the primary metric, average number of steps, and average dollar cost per task as secondary metrics. Cost is computed from token counts recorded by the LiteLLM callback multiplied by published model rates. Each task is hard-capped at 150 steps and at a per-task budget of \$1 (small models: Haiku 4.5, GPT-5.4-mini) or \$3 (large models: Sonnet 4.6, GPT-5.4). Tasks that exhaust either limit are counted as failures.

6.2 Are the wrappers faithful?

A wrapped benchmark is only useful if it is the same benchmark. We treat that as a measurement problem with four layers, each catching a different way a wrapper can silently diverge, all reproducible from `scripts/paper/`.

Layer 1: task-set identity. Every shipped task id and instruction string is compared against the upstream task index at the pinned commit. OSWorld matches on **368/368** ids with **368/368** byte-identical instructions; Windows Agent Arena on **152/152** and **152/152**. Agents read exactly what upstream specifies.

Layer 2: evaluation provenance. Scoring code reaches a CUBE by one of three routes, each admitting a different check (Table 4). Three CUBEs *delegate* scoring to the original authors’ published package (`browsergym-workarena`, `miniweb`, `webarena-verified`), so the evaluator is upstream’s by construction. Three run the benchmark’s *own image and test commands* (the official `swebench/sweb.eval.*` images; Terminal-Bench’s per-task images). The two CUA CUBEs *vendor* upstream evaluator source, which we audit function by function at the AST level, normalising edits that cannot change a score (docstrings, logging, annotations, lint-driven renames). Of 236

Table 2: Pass rate (%) per model and CUBE on the evaluation subsets. Best per row is shown in bold.

CUBE (subset)	# Tasks	Anthropic		OpenAI	
		Haiku 4.5	Sonnet 4.6	GPT-5.4-mini	GPT-5.4
WorkArena L1	33×10	31.7 ±2.6	65.6 ±2.6	27.3 ±2.5	53.6 ±2.7
WorkArena L2	235	15.6 ±2.4	35.5 ±3.1	5.2 ±1.4	35.7 ±3.1
WebArena-Verified	381	—	14.8 ±1.8	—	9.0 ±1.5
MiniWoB	125×5	67.2 ±1.9	59.4 ±2.0	52.5 ±2.0	52.5 ±2.0
OSWorld (PyAutoGUI)	360	43.2 ±2.6	49.4 ±2.6	17.3 ±2.0	28.6 ±2.4
OSWorld (Computer13)	360	42.1 ±2.6	51.6 ±2.6	27.5 ±2.4	30.7 ±2.4
Win. Agent Arena (PyAutoGUI)	154	41.5 ±4.0	61.2 ±3.9	25.7 ±3.5	37.5 ±3.9
Win. Agent Arena (Computer13)	154	32.9 ±3.8	32.2 ±3.8	23.7 ±3.4	27.0 ±3.6
SWE-bench Verified	500	43.0 ±2.2	65.1 ±2.1	45.4 ±2.2	58.2 ±2.2
SWE-bench Live (lite-gold)	223	20.2 ±2.7	38.6 ±3.3	22.3 ±2.8	35.4 ±3.2
TerminalBench	89	14.0 ±3.7	29.5 ±4.8	12.4 ±3.5	11.4 ±3.4

OSWorld evaluator functions **90.3 %** are semantically identical, with 13 possibly score-affecting; for Windows Agent Arena, **74.5 %** of 274, with 23. The residue is dominated by two mechanical patterns, extraction of upstream’s repeated per-OS path-resolution preamble into one helper and pruning of branches for operating systems the CUBE image never runs, plus the evaluator bug fixes the judge surfaced (§6.6). Every residual diff ships with the paper.

Layer 3: oracle parity. Does the evaluator award full credit to the benchmark’s own reference solution? Each SWE CUBE ships a scripted, LLM-free gold-patch agent; Terminal-Bench exposes the same check through `oracle_mode`, which uploads upstream’s own `solve.sh`. On SWE-bench Live’s 300-task lite split, **275** tasks resolve under the CUBE evaluator. The residue is why we report the check rather than assume it: an earlier run on non-root infrastructure resolved only 223, so roughly 62 tasks turn on whether the container runs as root, not on evaluation semantics. Agents are evaluated on the gold-solvable subset, which is what *lite-gold* in Table 2 means.

Layer 4: cross-harness verdict agreement. The sharpest test holds the artefact fixed and varies the grader. Episodes now record the agent’s model patch, so identical patches can be re-graded by the upstream SWE-bench harness and verdicts compared task by task. We expect a small, one-directional disagreement, because our evaluator makes three deliberate relaxations that we state rather than hide: PASS_TO_PASS tests already failing before the agent touches the repository are not charged to it, a fixed list of network-dependent tests is skipped offline, and `pytest`’s “no tests collected” is not a failure where upstream task data carries untokenisable test ids. Each avoids scoring a correct fix as 0 for reasons outside the agent’s control; each also makes our SWE numbers a possible upper bound against the official harness, which is precisely the kind of difference a parity check exists to quantify.

What the judge’s blame rate is, and is not. An earlier version of this work used the judge’s benchmark-side blame rate as the primary faithfulness signal. It is not one: it is the wrapper’s own instrument, and a low rate cannot certify the wrapper any more than a compiler can certify its own output. Its real role is *monitoring*, surfacing candidate wrapper defects at corpus scale, and it found real ones, including an escape-sequence bug in an evaluator path that suppressed five Windows Agent Arena scores (§6.6). Layers 1 to 4 establish faithfulness; the blame rate keeps finding the next defect.

6.3 Baseline pass rates

Table 2 reports per-model, per-CUBE pass rate across the subsets defined above. Sonnet 4.6 leads on most CUBEs, doubling the small models’ rates on the harder subsets (WorkArena L2, SWE-bench Live, TerminalBench); on easier or saturating subsets (WorkArena L1, MiniWoB) the gap closes and Haiku occasionally tops the column. SWE shows the steepest capability cliff: small models compress the SWE ranking sharply, suggesting long-horizon coding benefits disproportionately from capability rather than scaffolding. These are *platform measurements*, not SOTA claims: Genny is a single generalist with one prompt and one tool stack per modality, no per-benchmark tuning, and a flat \$1/\$3 budget cap. Bespoke pipelines that tune prompts and tools per benchmark routinely score higher; our goal is comparable numbers under a fixed configuration, not leaderboard positions.

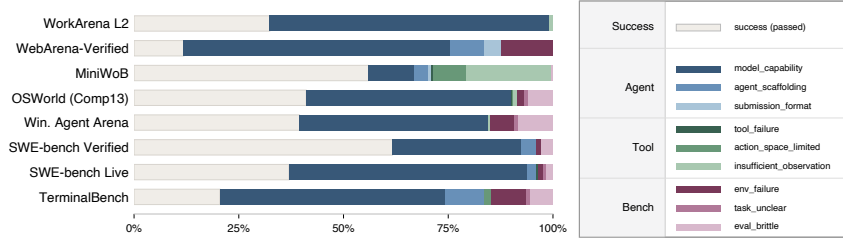


Figure 2: Blame distribution per benchmark. Each bar spans 100 % of tasks: the pale left segment is the pass rate (averaged over Sonnet 4.6 and GPT-5.4); the remainder is divided among the 9 failure-attribution categories (excluding none), scaled proportionally to the judge’s blame counts.

6.4 Where do failures originate?

Aggregating the structured judge outputs across the corpus answers both the accountability question and the faithfulness question. The dominant cause of failure is the agent itself: limitations in the underlying LLM’s reasoning, planning, or context management account for roughly **82 %** of attributions, with the remaining **18 %** split between benchmark-side (**11 %**) and tool-side (**6 %**) issues. Figure 2 breaks the distribution down by benchmark using the judge’s nine failure-attribution categories, grouped visually into agent-side (blue shades), tool-side (green), and benchmark-side (rose) buckets.

Two patterns stand out. On CUA and SWE the agent-side share exceeds 80 %, with `model_capability` alone accounting for roughly four-fifths of attributions: closing this gap is the LLM-research problem, not an evaluation one. MiniWoB is the single benchmark where tool-side blame dominates: nearly 65 % of its failures hit `action_space_limited` or `insufficient_observation` because the BrowserGym bid action set cannot express the pixel-level mouse interactions that roughly 30 % of MiniWoB tasks require, illustrating that tool configuration is an independent variable. The benchmark-side share concentrates on Windows Agent Arena and TerminalBench; each flagged case is a queued issue we are iterating on (Appendix H).

6.5 Does the picture survive a change of scaffold?

Table 2 measures one agent. Since two scaffolds calling the same model can differ substantially on the same benchmark (Appendix E), a single-scaffold table invites the question of whether it measures the models or the scaffold. We therefore ran a second, structurally different public scaffold over an identical task set with an identical tool stack and model: ReAct, a think/act transcript loop, against Genny’s rolling-summary context management, both on GPT-5.4-mini over all 125 MiniWoB tasks through the same BrowserGym action set (one seed, 10-step cap; this is a controlled scaffold comparison, not a re-run of Table 2’s configuration).

Aggregate pass rates are close: **66.1 %** (Genny) against **69.4 %** (ReAct) on the 124 tasks both completed, a **+3.2 pp** difference a paired exact McNemar test does not distinguish from zero ($p=0.42$). The task-level view is more informative: the scaffolds reach the *same* outcome on **88.7 %** of tasks (Spearman $\rho=0.74$), with 14 discordant tasks split 5 to 9. Roughly one MiniWoB task in nine is decided by the scaffold rather than the model, while the benchmark-level reading is stable across both. Holding the tool stack constant is what keeps that variance small: the two loops differ in context management but act through the identical action space, so neither can win by having better tools. It also bounds our claim. Table 2 reports platform measurements of one configuration; conclusions resting on gaps of a few points should not be drawn from a single-scaffold table, including ours.

6.6 Qualitative findings from the judge

Beyond the structured fields, the judge’s free-form *analysis* acts as a curated review of the corpus. Three recurring failure shapes show up repeatedly across our 8 CUBEs: capability gaps, where the relevant evidence is in the trajectory but the agent fails to connect it; scaffolding pathologies, typically deterministic loops or context exhaustion that the agent cannot escape on its own; and wrapper or evaluator bugs that only become visible once several episodes are aggregated. Appendix H works through one three-judge case per category. The benchmark-side case (VLC `v1crc` cluster, Table 10)

424 is representative: five disjoint episodes flagged the same Python escape-sequence trap, and a one-line
425 `os.path.join` fix flipped all five from 0.0 to 1.0 on resume.

426 7 Discussion

427 **Wrapping a benchmark is now a small, well-scoped job.** Under the standard, integrating a new
428 benchmark reduces to implementing a handful of typed classes (`Task`, `Benchmark`, `ToolConfig`)
429 plus declarative metadata, and inheriting the shared resource lifecycle, JSON-RPC interface, and
430 compliance test suite for free. In our own experience the per-benchmark wrapping cost collapsed
431 from weeks of bespoke harness integration to roughly a day of focused work. We further release a
432 `/new-cube` Claude skill that scaffolds the wrapper, generates the metadata, and runs the compliance
433 checks interactively; with this tool, a benchmark author who knows their own benchmark can produce
434 a passing CUBE in an afternoon. We expect this combination of a small, opinionated standard and
435 an LLM-driven wrapping assistant to be the dominant lever for growing the registry.

436 **The judge turns every run into a debugging signal.** A recurring pattern emerged across our 8
437 CUBEs: several independent failed episodes converge on the same root cause, and the fix is usually
438 a single line of wrapper or evaluator code (the VLC cluster in §6.6 is one such case). Beyond
439 attributing failures to the agent, the judge therefore functions as a continuous review of the wrappers,
440 and of the underlying benchmarks themselves. Several `should_have_been_rewarded` verdicts
441 in our runs pointed at genuine ambiguity in task descriptions or evaluator logic, artefacts that the
442 original benchmark authors are in a position to refine. A shared judge protocol thus turns evaluation
443 runs into a shared queue of triaged, evidence-backed issues for the entire benchmark ecosystem.

444 **Limitations.** Three bear directly on how far our claims should be read. *The RL post-training*
445 *motivation is untested end-to-end*: we motivate CUBE by multi-environment RL and demonstrate
446 evaluation of four closed API models. CUBE supplies what such a pipeline needs (common task
447 interface, resource lifecycle, trajectory export, NeMo Gym connector), and the episode loop that
448 feeds it sustains $\approx 1,200$ complete MiniWoB rollouts per hour on a single 8-core laptop with four
449 Ray workers, but this paper reports no training run and no open-weight model, so the RL argument is
450 a design argument, not a result. *Faithfulness is argued, not proven*: each layer in §6.2 has a boundary.
451 Task-set identity and provenance are static and cannot catch a runtime divergence; oracle parity covers
452 only benchmarks shipping a reference solution, excluding the CUA CUBEs; cross-harness agreement
453 exists only where a second harness does. For web and CUA we have provenance and task identity
454 but no independent second grader. *Scaffold and tool choices are part of every number*: §6.5 bounds
455 scaffold variance on one benchmark and does not license extrapolating that bound to the longer
456 horizons of CUA and SWE. Beyond these, streaming and multi-agent settings are not yet supported
457 and are the next protocol extensions we plan to ship; dataset-only benchmarks remain out of scope;
458 “tools held constant” is harness-enforced rather than standard-enforced; the compliance suite is a
459 floor, not a correctness guarantee; and all empirical claims extend only to the 8 CUBEs evaluated.

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619 paper’s contributions and scope?

620 Answer: [Yes]

621 Justification: §1 enumerates four concrete contributions (standard, harness, corpus and
622 registry, cross-benchmark experiments) and states the scope explicitly (10 CUBEs, three
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Justification: Table 2 reports pass rate ± 1 binomial standard error ($\sqrt{p(1-p)/n}$) for each (model, CUBE) pair, where n is the effective task count (tasks $\times$ seeds where multiple seeds are used). The metric is task-binary (pass/fail), so the binomial SE is the appropriate measure of task-level variance within each evaluation subset.

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Appendix

A JSON-RPC interface details

The standard auto-generates a JSON-RPC 2.0 server from any `Benchmark` subclass. The server exposes two endpoints whose names and payload shape match MCP’s `tools/list` and `tools/call`, so any MCP-compatible client (LLM tool-call loop, IDE plugin, evaluation harness) can drive a CUBE task without an adapter layer.

Endpoint: `tools/list`. Returns the action schema derived from the task’s default tool: one entry per `@tool_action`-decorated method, with name, description, and JSON Schema for the arguments. The schema is generated at task construction time and is static for the lifetime of the episode.

Endpoint: `tools/call`. Accepts `{name, arguments}` and routes to `task.step(Action(name, arguments))`, returning the serialized `EnvironmentOutput` (observation, done flag, reward if terminal, info dict). A call after `done=True` raises a protocol error; the client must call `tools/list` on a fresh session to start a new episode.

Server lifecycle. `Benchmark._setup()` starts the server (one per benchmark, shared across tasks); `Benchmark.close()` tears it down. Individual tasks register and deregister their endpoints on `reset()` and episode end. The server address is published in `RuntimeContext` so remote workers can connect without further coordination.

B Per-benchmark task count, license, infra requirements

Table 3: The CUBE registry at submission.

CUBE	Modality	# Tasks	Backend
WorkArena	Web	333	Live enterprise web app
WebArena-Verified	Web	812	Shared 6-service Docker stack
MiniWoB	Web	125	Local browser
BrowseComp	Web	1,266	Browser
OSWorld	CUA	368	Ubuntu 22.04 VM
Windows Agent Arena	CUA	154	Windows 11 VM
SWE-bench Verified	SWE	500	Per-task Docker
SWE-bench Live	SWE	1,895	Per-task Docker
Terminal-Bench	Terminal	89	Per-task Docker
DRBench	Research	100	Per-task Docker (Nextcloud, Mattermost, IMAP)
Total		5,642	

Table 4: Static parity audit of the corpus (`scripts/paper/parity_audit.py`). Task-set match compares shipped task ids and instruction strings against the upstream index at the pinned commit. Evaluator match is AST-level semantic identity of upstream evaluator functions, ignoring edits that cannot change a score.

CUBE	Evaluation provenance	Task-set match	Evaluator match
WorkArena	upstream package	—	<code>browsergym-workarena</code>
WebArena-Verified	upstream package	—	<code>webarena-verified</code>
MiniWoB	upstream package	—	<code>miniwob</code>
OSWorld	vendored source	368/368	213/236 fn (90.3 %)
Windows Agent Arena	vendored source	152/152	204/274 fn (74.5 %)
SWE-bench Verified	upstream image + tests	500/500	official image, 3 documented relaxations
SWE-bench Live	upstream image + tests	300/300 (lite)	275 gold-solvable (oracle)
Terminal-Bench	upstream image + tests	89/89	upstream <code>solve.sh</code> oracle

C Wrapping effort

Section 7 claims that wrapping collapses to a handful of typed classes. Table 5 measures it (scripts/paper/wrapping_effort.py), in non-blank, non-comment, non-docstring lines. Vended upstream evaluator source is excluded from the wrapper column: it is the original benchmark’s own scoring code, not authoring effort, and its provenance is audited separately in §6.2. The debug suite is broken out because it is the one artefact the standard requires beyond the wrapper itself, and it is what the registry’s compliance check runs.

The median CUBE is **508** lines of authored wrapper plus a 33 to 144 line debug suite. The spread is informative: the cheapest wrappers are the ones whose upstream ships an importable evaluation package (MiniWoB, WorkArena), and the most expensive are those that must reconstruct a provisioning path the upstream project only expressed as a run script (OSWorld, Terminal-Bench).

Table 5: Authored wrapping effort per CUBE, in SLOC.

CUBE	Wrapper	Debug suite	Vended evaluator
WorkArena	325	33	—
WebArena-Verified	508	37	—
MiniWoB	264	67	—
BrowseComp	253	113	—
OSWorld	798	144	8,225
Windows Agent Arena	571	85	8,771
SWE-bench Verified	460	38	—
SWE-bench Live	561	38	—
Terminal-Bench	593	73	—
Median	508	67	

D Judge output schema and blame taxonomy

Output schema. Each JudgeOutput carries five fields: an *outcome* (success, success_lucky, almost, failure, should_have_been_rewarded); a *primary blame* from the closed taxonomy plus optional *secondary blames*; *evidence*, the step-indexed verbatim quotes; a *hypothesis* of what would fix the failure class; and 0–5 confidence scores on both blame and hypothesis. A free-form *analysis* field placed first acts as a reasoning scratchpad before the structured fields are committed.

Blame taxonomy. Ten benchmark-agnostic categories grouped into three buckets. **Agent-side:** model_capability (the LLM lacked the reasoning, knowledge, or planning to solve the task); agent_scaffolding (the agent loop, prompt, or context management caused the failure); submission_format (the agent reached a solution but failed to submit it through the registered channel). **Tool-side:** tool_failure (a tool wrapper raised or returned an unexpected error); action_space_limited (no available action could express the correct solution); insufficient_observation (the rendered observation hid information the agent needed). **Benchmark-side:** env_failure (infrastructure crash outside the agent’s and tool’s control); task_unclear (the task description is ambiguous or contradictory); eval_brittle (the evaluator rejected an acceptable solution). A residual none category is assigned on clean success or when evidence is insufficient for any attribution.

E Drift sources in dynamic-agent benchmarks

Five sources of drift recur across the corpus and motivate the reliability framing of Appendix E.

Scaffolding sensitivity. Pass rate depends heavily on the agent loop, the tool wrappers, the prompt boilerplate, and the exact context-management strategy. Two scaffolds calling the same model can differ by tens of points on the same benchmark; cross-benchmark comparisons that mix scaffolds are not measuring the model.

Prompt tailoring. Many published numbers use a prompt or chain-of-thought structure tuned for a specific benchmark, sometimes with in-context examples drawn from its training split. The reported

number is then a joint property of the model and the prompt-engineering effort, not of the model in a generalist setting.

Model version drift. Frontier APIs are continuously retrained and behaviour-tuned. A model identifier (claude-sonnet-4-6, gpt-5.4) names a moving target whose behaviour at the time of a published number may not be reproducible months later.

Closed-source scaffolds. Strong leaderboard numbers are sometimes reported by labs whose scaffold, tool stack, and prompt are not released, in contexts where headline numbers carry commercial weight. These results are not reproducible by construction.

Live-component drift. Several benchmarks in the corpus depend on live external services or system state that change without notice: WorkArena targets a live enterprise web application; OSWorld and Windows Agent Arena depend on OS revisions; underlying browser-automation libraries (playwright, pyautogui) ship breaking updates between minor versions. Two runs weeks apart against the same task set can produce different ground-truth pass rates.

F Judge meta-analysis

For every episode that received three independent judge invocations we measure *primary-blame agreement* (do all three judges choose the same blame category?) and *outcome agreement* (do all three agree on pass / almost / fail?). Tables 6 and 7 report the counts and rates for the two benchmarks where triple-judge runs were completed: SWE-bench Verified (Sonnet 4.6, $n=162$; GPT-5.4, $n=204$) and SWE-bench Live (Sonnet 4.6, $n=83$).

Table 6: Primary-blame agreement across three independent judges. 3/3: all three choose the same category. 2/3: a two-judge majority. All differ: no two judges agree. $\geq 2/3$ subsumes both 3/3 and 2/3 rows.

Benchmark	Model	n	3/3	2/3	All differ	3/3 %	$\geq 2/3$ %
SWE-bench Verified	Sonnet 4.6	162	117	40	5	72.2	96.9
SWE-bench Verified	GPT-5.4	204	173	27	4	84.8	98.0
SWE-bench Live	Sonnet 4.6	83	64	19	0	77.1	100.0

Table 7: Outcome agreement (pass / almost / fail) across three independent judges. Column definitions identical to Table 6.

Benchmark	Model	n	3/3	2/3	All differ	3/3 %	$\geq 2/3$ %
SWE-bench Verified	Sonnet 4.6	162	114	46	2	70.4	98.8
SWE-bench Verified	GPT-5.4	204	126	77	1	61.8	99.5
SWE-bench Live	Sonnet 4.6	83	77	6	0	92.8	100.0

What this does and does not show. Agreement is a measure of *consistency*, not of *correctness*: three judges can agree and all be wrong, and nothing in this table rules that out. Appendix G reports the two checks that do bear on correctness, namely agreement against human labels and against a judge running on a different model family.

Takeaways. Majority ($\geq 2/3$) agreement is **97–100 %** across every condition for both blame and outcome, meaning the modal vote is almost always stable and reliable as a signal. Full unanimous agreement is somewhat lower — **62–85 %** for blame and **62–93 %** for outcome — which is expected: the taxonomy contains adjacent categories (*model_capability* vs. *agent_scaffolding*) that can legitimately both apply to the same failure, and the outcome three-way split (pass / almost / fail) places hard boundaries on a continuous quality gradient.

The hardest condition is SWE-bench Verified outcome for GPT-5.4, where unanimous agreement falls to 61.8 %. We attribute this to the high overall pass rate on that benchmark ($\approx 58\%$): many GPT-5.4 episodes produce partial fixes that sit ambiguously between *almost* and *fail*, a genuinely

difficult judgement call even for human annotators. Notably, the *all-differ* rate (complete three-way disagreement) stays below 2.5 % in every condition, confirming that true confusion is rare and that majority voting is a robust aggregation strategy.

G Judge validation against humans and across model families

Appendix F establishes that the judge is stable. Stability is necessary and not sufficient, so we run two further checks. Both are released as tooling (`scripts/paper/judge_validation.py`) rather than one-off analysis, because a judge protocol that other groups adopt should ship with the means to re-validate it on their own corpora.

Human gold labels. We draw a stratified sample of failed episodes, balanced over (modality, judge blame category) with a floor per cell so that the rare benchmark-side categories, which carry the faithfulness argument, are estimable at all rather than swamped by `model_capability`. Annotators receive a blind packet: the same task description and step-indexed trajectory the judge read, the identical closed taxonomy, and no sight of the judge’s verdict, which removes the anchoring that makes post-hoc “does this look right?” review worthless. Each episode is labelled independently by at least two annotators. We report Cohen’s κ between the human consensus and the judge on the 10-category taxonomy and on the three-bucket grouping (agent / tool / benchmark), plus inter-annotator κ as the ceiling any automated judge could reach on this task. The bucket-level number is the one the paper’s conclusions rest on: confusing `model_capability` with `agent_scaffolding` barely moves the accountability picture, whereas confusing agent-side with benchmark-side moves it entirely.

Swapping the judge’s backbone. A judge that only ever runs on one model family cannot separate what the trajectory shows from what that family believes about trajectories. The investigator’s driver is a Protocol, so the backbone is a call-time argument: alongside the two Claude drivers we ship `CodexDriver`, which runs the identical recipe, prompts, and output schema through `codex exec` on a non-Anthropic model. Re-judging the same episodes on both backbones yields per-episode agreement and, separately, the shift in the aggregate blame distribution. The distinction matters: the corpus-level claim (§6.4) needs the *distribution* to be stable, and a judge pair can disagree on individual hard episodes while agreeing on the shape of the corpus.

In a pilot on six MiniWoB episodes, a Sonnet-class judge and a GPT-5.4 judge selected the same outcome and the same primary blame on every episode ($\kappa=1.0$), including the two episodes both attributed to `action_space_limited` rather than to the model. Six short episodes from one benchmark cannot bound cross-backbone disagreement and we do not present this as an estimate: MiniWoB trajectories are brief and unusually legible, which is the easy end of the judging task. It establishes that the comparison runs end to end and that the closed taxonomy transfers across model families without prompt surgery. The corpus-scale swap is the number that belongs in this table.

Reproducing these checks.

```
# 1. human study
judge_validation.py sample <results> --n 100 --out study/
judge_validation.py packet study/ --annotator <name>
judge_validation.py score study/ --labels study/labels-*.json

# 2. backbone swap
judge_validation.py snapshot <results> --label claude --out claude.json
ch-investigate run <results> --driver codex --overwrite
judge_validation.py snapshot <results> --label codex --out codex.json
judge_validation.py swap claude.json codex.json
```

H Judge case studies

Table 8 shows a representative case in which all three judges unanimously converge, illustrating both the evidence-grounding mechanism and the blame attribution in practice. The episode is drawn from the SWE-bench Verified run (§6); the judge had access to the full step-indexed trajectory and the privileged evaluation output.

Table 8: Three independent judge outputs for episode `astropy__astropy-14365` (SWE-bench Verified, outcome *almost*). All three judges unanimously agree on primary blame and intervention; high-confidence evidence quotes pinpoint the unfixed downstream `if v == "NO":` comparison that the agent read but failed to connect to its own regex fix. Evidence quotes are verbatim from the judge output; long code blocks are truncated at [...] to fit the column.

Field	Judge 1	Judge 2	Judge 3
Outcome	<i>almost</i>	<i>almost</i>	<i>almost</i>
Primary blame	<code>model_capability</code> (conf. 4/5)	<code>model_capability</code> (conf. 3/5)	<code>model_capability</code> (conf. 4/5)
Secondary	<code>agent_scaffolding</code>	–	<code>agent_scaffolding</code>
Summary	Added <code>re.IGNORECASE</code> to QDP command regex but missed downstream <code>if v == "NO":</code> null-marker check; verified only syntax compilation, never ran the test suite.	Fix incomplete: <code>re.IGNORECASE</code> applied to command recognition; null-value sentinel "NO" still compared case-sensitively; crashes on lowercase "no" with <code>ValueError</code> .	Regex fix correct for command recognition; identical case-sensitivity bug in data-value processing loop not connected; <code>final_step</code> called without running any tests.
Key evidence	<code>step 8 (verified):</code> <code>for v in</code> <code>line.split(delimiter):</code> <code>if v == "NO":</code> <code>...</code> <code>else: [...]</code> <code>step 16 (verified):</code> <code>ValueError: could not</code> <code>convert string to float:</code> <code>'no'</code>	<code>step 56 (verified, match 0.88):</code> <code>ValueError: could not</code> <code>convert string to float:</code> <code>'no'</code> <code>astropy/io/ascii/qdp.py:317:</code> <code>ValueError</code>	<code>step 8 (verified):</code> <code>for v in</code> <code>line.split(delimiter):</code> <code>if v == "NO":</code> <code>...</code> <code>else: [...]</code> <code>step 15 (verified):</code> <code>ACTION final_step:</code> <code>{}</code>
Hypothesis	Change "NO" to <code>v.upper() == "NO"</code> throughout; mandate <code>pytest</code> run on the failing test before submit.	Run the actual failing test before submitting; would reveal the secondary failure immediately.	Add mandatory post-fix validation step (run failing reproduction case) to the agent workflow.
Intervention	<code>prompt_change</code> (conf. 3/5)	<code>prompt_change</code> (conf. 3/5)	<code>prompt_change</code> (conf. 4/5)

Table 9: Three independent judge outputs for episode `conan-io__conan-18028` (SWE-bench Live, outcome *failure*). The agent located `graph_builder.py` and the `skip_test` check at step 82, then immediately fell into a deterministic loop, re-issuing the same `grep` command 53 consecutive times and hitting `MAX_STEPS_REACHED` without writing a single line of code. All three judges agree on `scaffold_change` as the intervention; two of three assign primary blame to `agent_scaffolding`. Evidence quotes are verbatim from the judge output; long action strings are truncated at [...] to fit the column.

Field	Judge 1	Judge 2	Judge 3
Outcome	<i>failure</i>	<i>failure</i>	<i>failure</i>
Primary blame	<code>agent_scaffolding</code> (conf. 5/5)	<code>model_capability</code> (conf. 4/5)	<code>agent_scaffolding</code> (conf. 4/5)
Secondary	<code>model_capability</code>	<code>agent_scaffolding</code>	<code>model_capability</code>
Summary	Spent all 150 steps in read-only exploration, running identical <code>grep</code> commands dozens of times without writing a single code edit or calling <code>final_step</code> .	Spent all 150 steps in <code>grep/cat</code> exploration; from step 80 fell into a tight loop re-running the same <code>grep</code> with identical output each time.	Located <code>tools.graph:skip_test</code> in <code>graph_builder.py</code> at step 81, then immediately fell into a deterministic loop: same <code>grep</code> command 53 consecutive times, 0 code edits, <code>MAX_STEPS_REACHED</code> .
Key evidence	<i>steps 99, 101, 103</i> (verified, match 0.86): ACTION bash: {... <code>grep -rn "skip_test tools.graph:skip..."</code> <code>graph_builder.py</code>} <i>steps 295, 297, 299</i> (verified, match 0.90): ACTION bash: {... <code>grep -rn "nodetest\b" conans/conan/</code> <code>-include="*.py" grep -v "test/"</code>}	<i>step 81</i> (verified, match 0.86): ACTION bash: {... <code>grep -rn "skip_test tools.graph:skip..."</code> <code>graph_builder.py</code>} <i>step 121</i> (verified, match 0.86): ACTION bash: {... <code>grep -rn "skip_test tools.graph:skip..."</code> <code>graph_builder.py</code>}	<i>step 82</i> (verified): 216: <code>skip_test = node.conanfile.conf.get("tools.graph:skip_test",...)</code> 222: <code>if skip_test and require.test:</code> <i>step 83 → 151</i> (loop begins): ACTION bash: {... <code>grep -rn "skip_test tools.graph:skip_test..."</code> <code>graph_builder.py</code>} <i>metadata</i> (step 0): "status": "MAX_STEPS_REACHED", "prompt_tokens": 6430181, "cached_tokens": 6267737
Hypothesis	Adding loop detection (block on N consecutive identical tool calls) and context compaction at a reasonable threshold would prevent this class of failure.	A scaffold with loop detection would have interrupted the cycle and forced a different action; the agent had located the relevant code but lacked the push to synthesize an edit.	Full history, temperature 0, and no compaction create a deterministic feedback loop once repetition starts; aborting after K consecutive identical actions would break the cycle.
Intervention	<code>scaffold_change</code> (conf. 5/5)	<code>scaffold_change</code> (conf. 3/5)	<code>scaffold_change</code> (conf. 5/5)

Table 10: Three independent judges on three WAA episodes (ep87, ep88, ep151; all VLC-domain, all scored 0.0), working on disjoint trajectories with no shared context. All three converge on the same eval-side root cause: the `vlcrc` path string passed to `execute_python_command` is re-parsed as Python source, silently collapsing `\v` to `U+000B` (vertical tab) and corrupting the path before `get_file()` sees it. A one-line `os.path.join` fix on the VM side flipped all five affected episodes from 0.0 to 1.0 on a single resumed run.

Field	Judge / ep87	Judge / ep88	Judge / ep151
Outcome	<i>should_have_been_rewarded</i>	<i>should_have_been_rewarded</i>	<i>should_have_been_rewarded</i>
Primary blame	<code>eval_brittle</code> (conf. 3/5)	<code>eval_brittle</code> (conf. 3/5)	<code>tool_failure</code> (conf. 2/5)
Task	Set VLC recording folder to Downloads	Set VLC recording folder to Desktop	Disable “Resize interface to video size” in VLC
Summary	Agent completed navigation (Preferences → [...] → Downloads → Save). Evaluator’s <code>get_vlc_config</code> path literal re-parsed as Python; <code>\v</code> → <code>U+000B</code> corrupted <code>vlcrc</code> to <code>\u000blcrc</code> ; <code>get_file 500</code> , false <code>FileNotFoundError</code> .	Agent navigated Preferences and set Desktop as recording folder and saved. Evaluator’s <code>get_vlc_config</code> path literal re-parsed as Python; <code>\v</code> → <code>U+000B</code> corrupted <code>vlcrc</code> to <code>\u000blcrc</code> ; <code>get_file 500</code> , score 0.0.	Agent navigated Preferences and unchecked ‘Resize interface to video size.’ Evaluator’s <code>get_vlc_config</code> path literal re-parsed as Python; <code>\v</code> → <code>U+000B</code> corrupted <code>vlcrc</code> to <code>\u000blcrc</code> ; <code>get_file 500</code> , false <code>FileNotFoundError</code> .
Key evidence	<i>step 9</i> (verified): [ERROR] <code>get_file()</code> -- C:\Users\Docke\AppData\Roaming\u000blc\u000blcrc	<i>step 9</i> (verified): [ERROR] <code>get_file()</code> -- C:\Users\Docke\AppData\Roaming\u000blc\u000blcrc	<i>step 8</i> (verified): ACTION run_pyautogui: {“code”: “pyautogui.click(784, 689) # Save”} <i>step 9</i> (verified): [ERROR] <code>get_file()</code> -- Roaming\u000blc\u000blcrc
Hypothesis	Path literal <code>vlcrc</code> in <code>execute_python_command</code> re-parsed as Python source; <code>\v</code> → <code>U+000B</code> . Replace with <code>os.path.join</code> to eliminate escape-sequence literals.	Classic escape-sequence trap on the Python round-trip. Build path from <code>os.path.join</code> components so no <code>\v</code> literal survives re-parse.	<code>get_vlc_config</code> sends a Python expression to the VM; <code>\v</code> in <code>vlcrc</code> collapses to <code>U+000B</code> silently. Switch to <code>os.path.join('~', ..., 'vlcrc')</code> .
Intervention	<code>eval_fix</code> (conf. 4/5)	<code>eval_fix</code> (conf. 4/5)	<code>eval_fix</code> (conf. 3/5)
Post-fix outcome	0.0 → 1.0 on resume (<code>is_vlc_recordings_folder</code> score 1.0)	0.0 → 1.0 on resume (<code>is_vlc_recordings_folder</code> score 1.0)	0.0 → 1.0 on resume (<code>is_vlc_resize_video_disabled</code> score 1.0)

Table 11: Three independent judge outputs for episode 716a6079_ep235 (**GPT-5.4 run**, OSWorld, outcome consensus: *failure*). Task: find `secret.docx` on the VM and copy its full path to the clipboard. File is at `/home/user/Data3/List3/secret.docx`. The three judges *disagree* on root cause: Judge 1 blames the model (agent used Ctrl+L in Nautilus, which copies the containing *directory* path, not the full file path, and rates the outcome *almost* — a near-miss sub-category of failure); Judges 2–3 blame infrastructure (`xsel` is not installed on the VM, so the evaluator `is_in_vm_clickboard` always reads an empty clipboard regardless of agent behavior). This split illustrates a compound failure: a model-capability weakness *and* a broken infrastructure-level evaluator co-occur, making the true verdict ambiguous. Compare with Table 12 (Sonnet run, same task): Sonnet used `xclip` correctly and received a unanimous *should_have_been_rewarded*.

Field	Judge 1	Judge 2	Judge 3
Outcome	<i>almost</i>	<i>failure</i>	<i>failure</i>
Primary blame	model_capability (conf. 3/5)	eval_brittle (conf. 4/5)	env_failure (conf. 4/5)
Secondary	action_space_limited	model_capability	model_capability
Summary	Agent found the correct file but used Ctrl+L+Ctrl+C in Nautilus, which copies the <i>directory</i> path, not the full file path. Clipboard contained <code>/home/user/Data3/List3</code> instead of <code>.../secret.docx</code> .	<code>xsel</code> is not installed on the VM (returncode 127), making clipboard scoring impossible for any agent. Separately, agent’s Ctrl+L+Ctrl+C would have produced the directory path, not the file path.	VM missing <code>xsel</code> binary that <code>is_in_vm_clickboard</code> relies on; evaluator always returns empty output, making reward=0 inevitable regardless of agent behavior.
Key evidence	<i>step 29</i> (verified, match 0.94): ACTION hotkey: {"keys": "ctrl+l"} <i>step 31</i> (verified, match 0.94): ACTION hotkey: {"keys": "ctrl+c"} <i>task config</i>: "expected": "/home/user/Data3/List3/secret.docx"	<i>evaluator log</i> (match 0.34): {'error': '/bin/sh: 1: xsel: not found\n', 'output': ", 'returncode': 127} <i>evaluator log</i> (match 0.52): reward=0.000000, evaluator=is_in_vm_clickboard <i>step 29</i> (verified, match 0.94): ACTION hotkey: {"keys": "ctrl+l"} "	<i>evaluator log</i>: {'error': '/bin/sh: 1: xsel: not found\n', 'returncode': 127, 'output': "} <i>evaluator</i>: is_in_vm_clickboard depends on xsel -clipboard -output
Hypothesis	Instruct agent to use file Properties dialog or right-click ‘Copy Location’ to obtain full file path; Ctrl+L in Nautilus yields only the directory.	Install <code>xsel</code> on VM base image, or update evaluator to fall back to <code>xclip -o -selection clipboard</code> ; also fix agent clipboard strategy.	VM image (cube-ubuntu-22-04 v1.0.0) does not include <code>xsel</code> ; install <code>xsel</code> or switch evaluator to <code>xclip -o or wl-paste</code> to fix all clipboard-check tasks.
Intervention	prompt_change (conf. 3/5)	eval_fix (conf. 5/5)	infra_fix (conf. 5/5)

Table 12: Three independent judge outputs for episode 716a6079_ep235 (**Sonnet run**, OS-World, outcome consensus: *should_have_been_rewarded*). *Same task as Table 11*. Unlike GPT-5.4 (which used Ctrl+L in Nautilus and copied only the directory path), the Sonnet agent correctly used `xclip -selection clipboard` in a terminal to copy the full file path `/home/user/Data3/List3/secret.docx`. All three judges agree: the correct agent action was irrelevant because the evaluator’s clipboard-reading command (`xsel -clipboard -output`) is not installed on the VM, so the clipboard always reads as empty. This cross-model pair is direct evidence that the same broken evaluator can produce a *model-capability failure* verdict for a weaker agent and a *should_have_been_rewarded* verdict for a stronger one — masking genuine capability differences behind infrastructure noise.

Field	Judge 1	Judge 2	Judge 3
Outcome	<i>should_have_been_rewarded</i>	<i>should_have_been_rewarded</i>	<i>should_have_been_rewarded</i>
Primary blame	eval_brittle (conf. 4/5)	eval_brittle (conf. 4/5)	eval_brittle (conf. 4/5)
Secondary	—	model_capability	env_failure
Summary	Agent correctly found the file and used <code>xclip</code> to copy its full path to the clipboard. Evaluator’s <code>xsel</code> getter is not installed on the VM (returncode 127), so it always reads an empty clipboard and returns reward=0.	Agent correctly copied the file path using <code>xclip -selection clipboard</code> , then called <code>done()</code> . Evaluator failed because <code>xsel</code> is not installed on the VM (returncode 127).	Agent correctly found <code>secret.docx</code> and copied its path via <code>xclip</code> . Evaluator reads clipboard with <code>xsel</code> , which is not installed (returncode 127), so comparison always sees an empty string.
Key evidence	<i>evaluator log</i> (match 0.31): <pre>{'error': '/bin/sh: 1: xsel: not found\n', 'returncode': 127, 'output': ''}</pre> <i>agent reasoning, step 9</i>: “The path <code>/home/user/Data3/List3/secret.docx</code> has been copied to the clipboard using <code>xclip -selection clipboard</code>.”	<i>agent reasoning, step 7</i>: “The file has been found at <code>/home/user/Data3/List3/secret</code> <i>evaluator log</i> (match 0.31): <pre>{'error': '/bin/sh: 1: xsel: not found\n', 'returncode': 127}</pre>	<i>step 15</i> (match 0.92): ACTION typing: <pre>{"text": "echo -n \" /home/user/Data3/List3/secret.docx\" xclip -selection clipboard"}</pre> <i>evaluator log</i>: <pre>xsel: not found, returncode 127</pre>
Hypothesis	VM image does not have <code>xsel</code> installed; install <code>xsel</code> or update evaluator to use <code>xdotool getclipboard</code> or <code>xclip -o</code> .	Install <code>xsel</code> on VM base image, or update evaluator to fall back to <code>xclip -o -selection clipboard</code> .	VM has <code>xclip</code> (installed) but not <code>xsel</code> ; evaluator’s getter always fails; fix evaluator to use <code>xclip -o -selection clipboard</code> .
Intervention	infra_fix (conf. 5/5)	infra_fix (conf. 5/5)	eval_fix (conf. 5/5)